

CLASH OF THE **CLAWS**

MSRIT_RootCause **CAPE:Context Aware** **Policy Engine**

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Problem Statement

Phones stay reactive while our day keeps changing



80+

unlock events
create repeated
context resets



23m

average refocus
penalty after
interruption



6x

daily mode changes
users manage
manually

Attention Breakdown



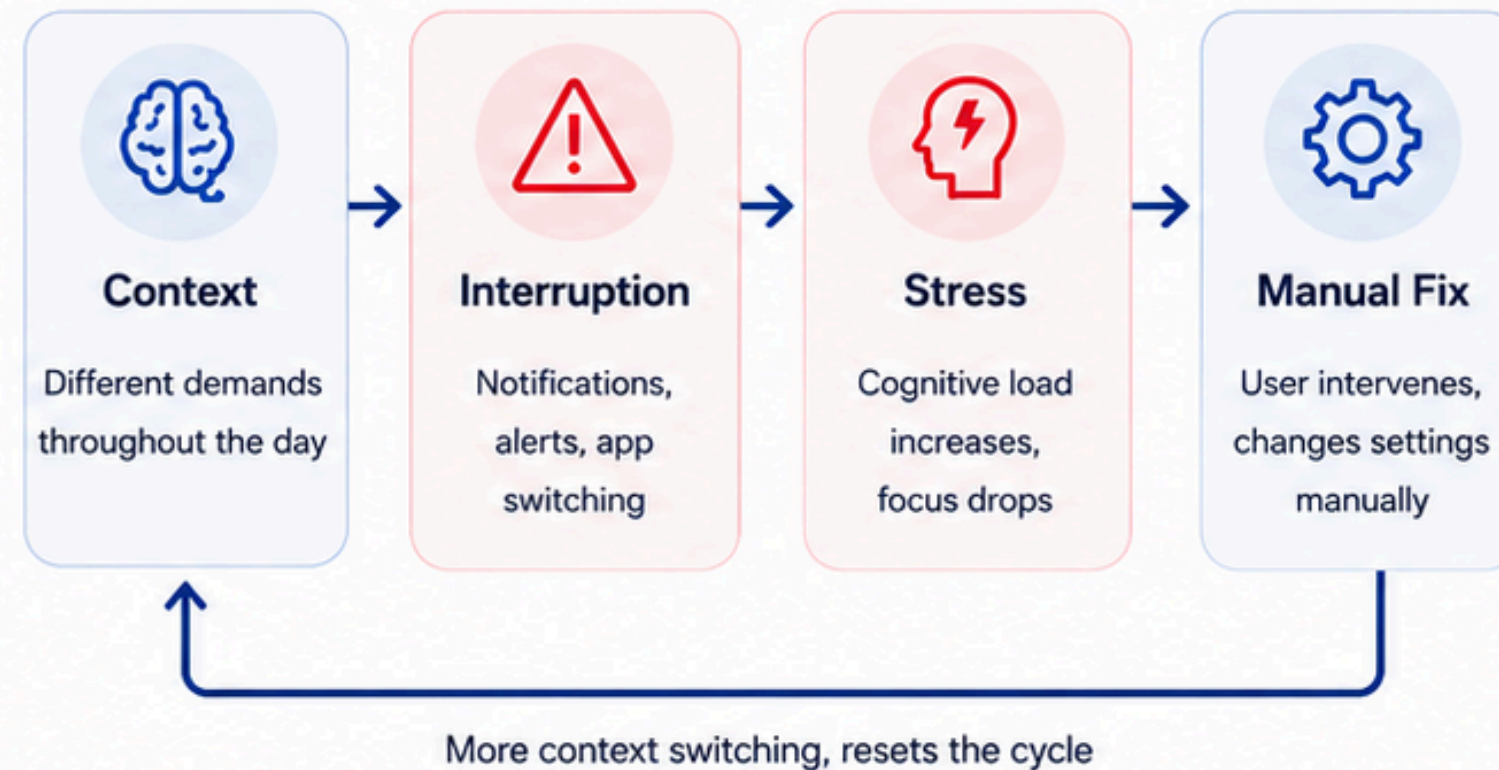
- High alert density trains users to ignore signals
- App switching fragments focus
- Static settings force frequent mode changes

Stress Accumulation



Sleep debt, unlock bursts, screen intensity and meeting pressure are observable proxies, but today they rarely influence automation.

Cognitive Overload Loop



Behavioral Signals We See



Notification density



High



App switching



Medium



Commute pressure

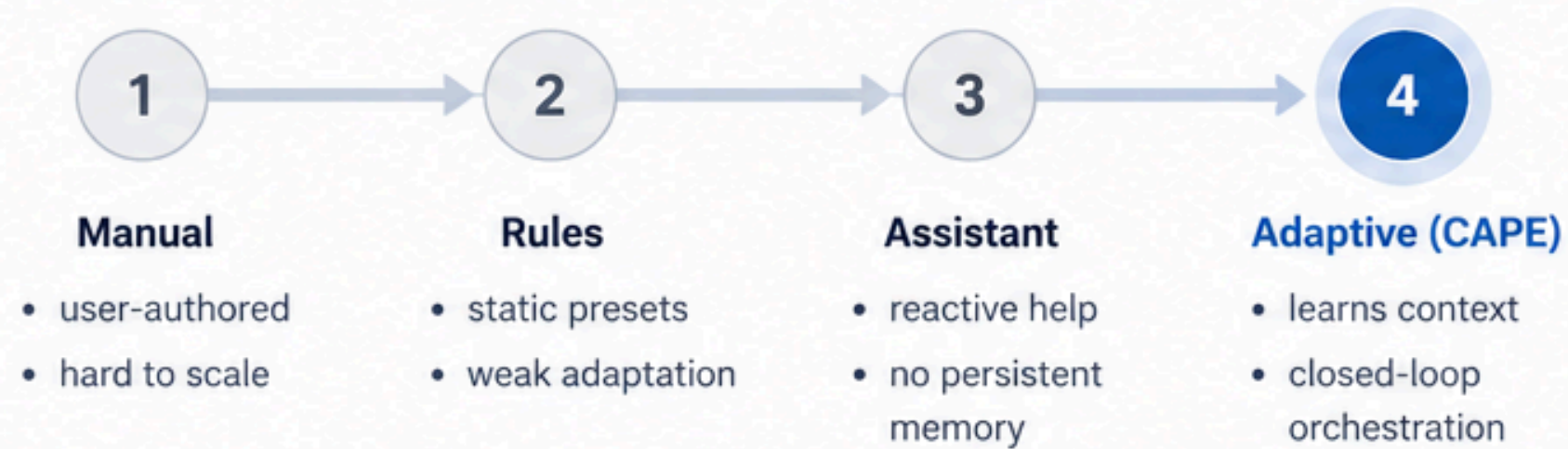


Medium

Signals are strong proxies for rising cognitive load and potential stress.

Current Solutions & Gaps

Automation intelligence maturity



WHY CURRENT SYSTEMS FAIL



Automate actions, not human state



No stress awareness



No confidence gating



No adaptive memory



STRATEGIC GAP

CAPE positions the smartphone as an adaptive policy platform, not a collection of one-off routines. It is the first system to combine real-time stress scoring, commute intelligence and override learning in one agent pipeline.

Capability benchmark

Static rules | Context fusion | Stress intelligence | Learning memory | Explainability



Modes

Good for predictable routines; weak when stress, commute or meeting pressure changes unexpectedly.



Tasker

Automation depth is high, but the burden of authoring and debugging rules stays with the user.



Assistant

Voice-first help solves tasks on demand, but lacks continuous context orchestration.



CAPE

Closed-loop intelligence selects behavior packs from live context, confidence and memory.



Context Fusion

Current tools



CAPE (Adaptive Intelligence)



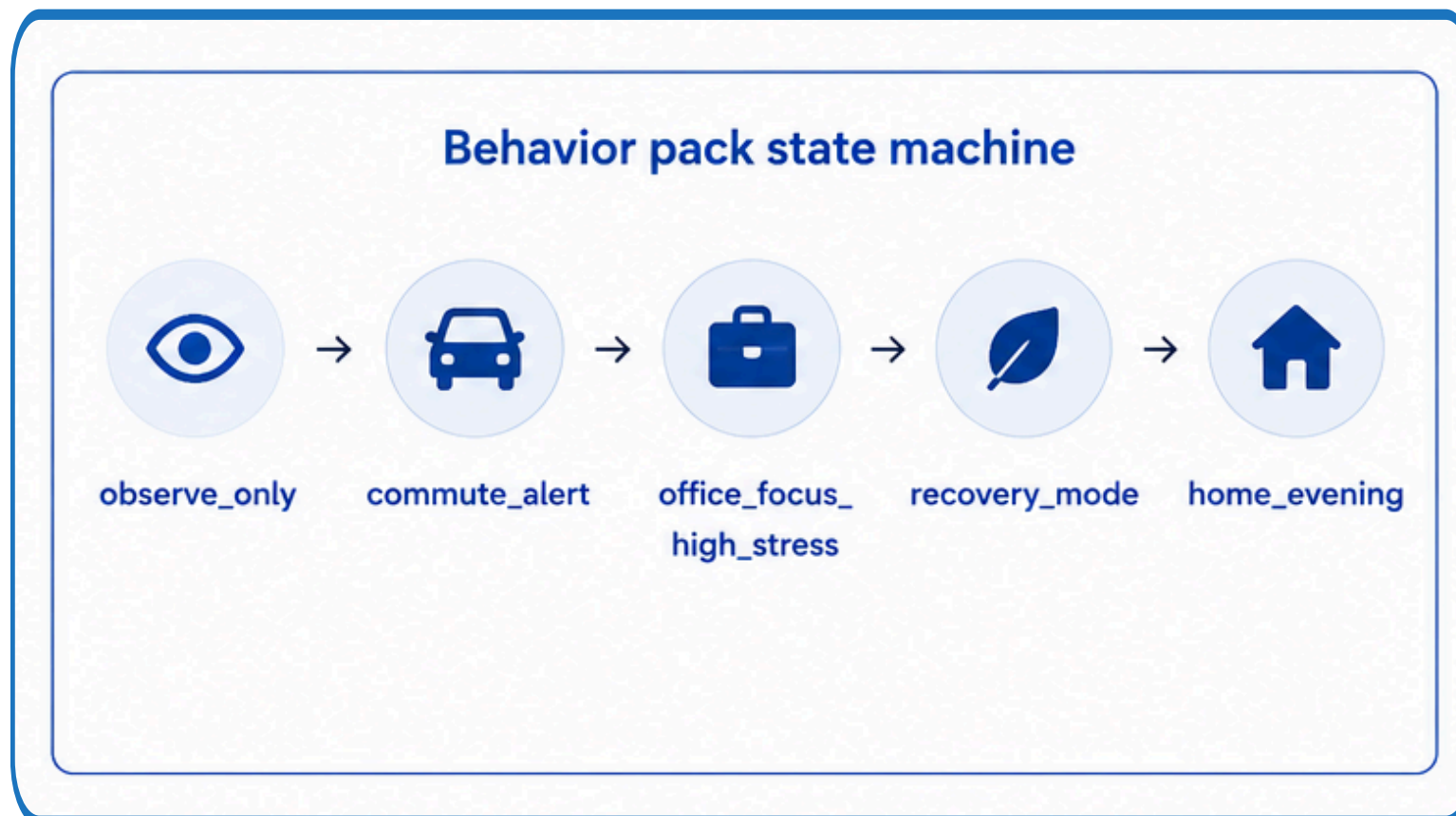
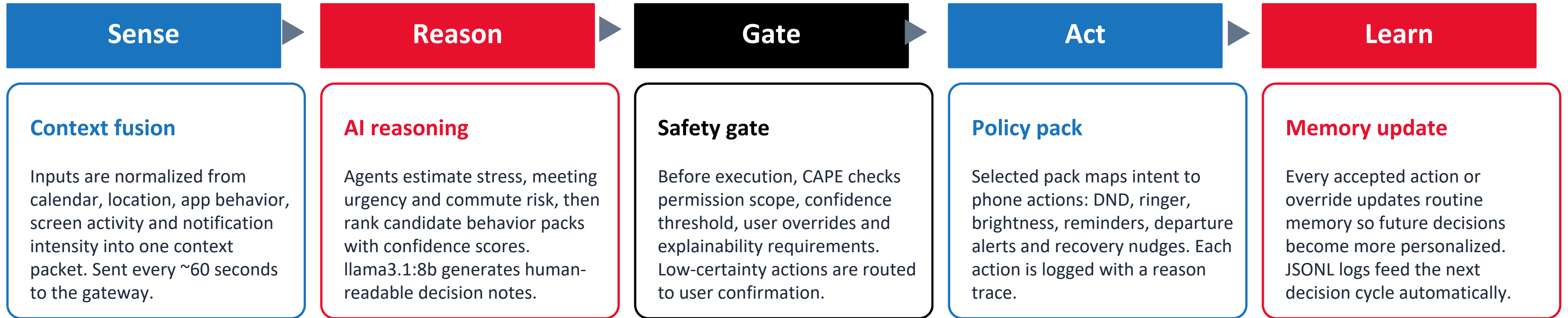
Stress Intelligence



Learning Memory



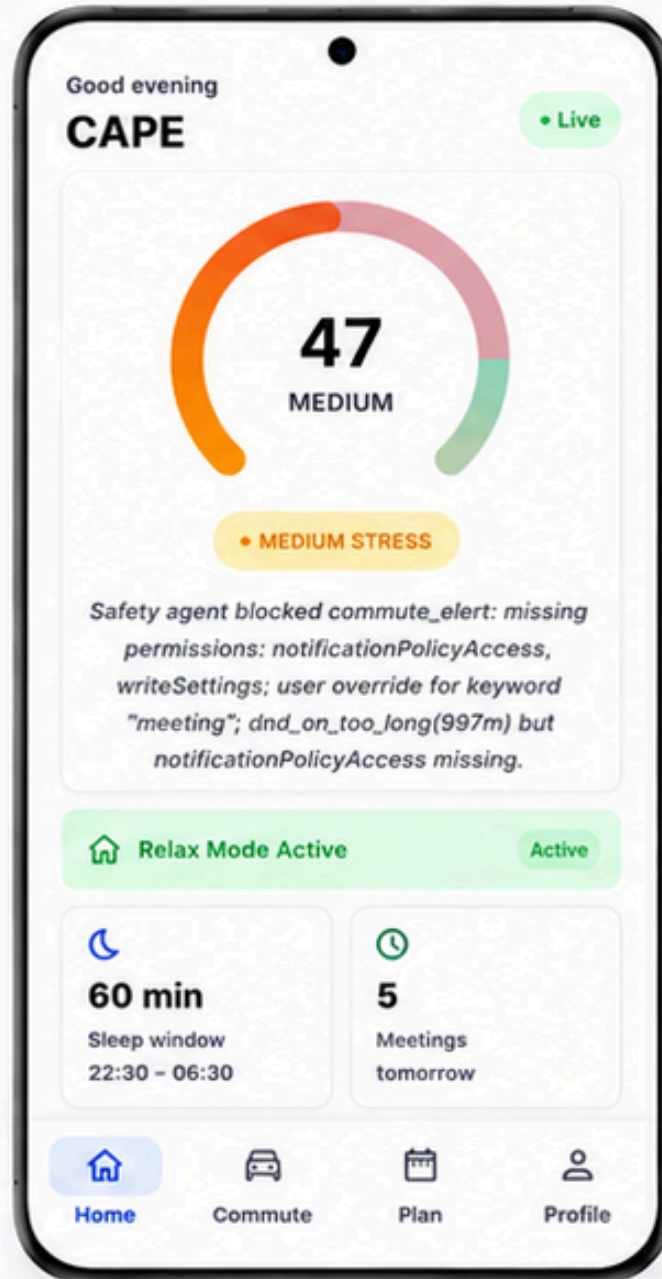
Our Solution



Demo/Product Walkthrough

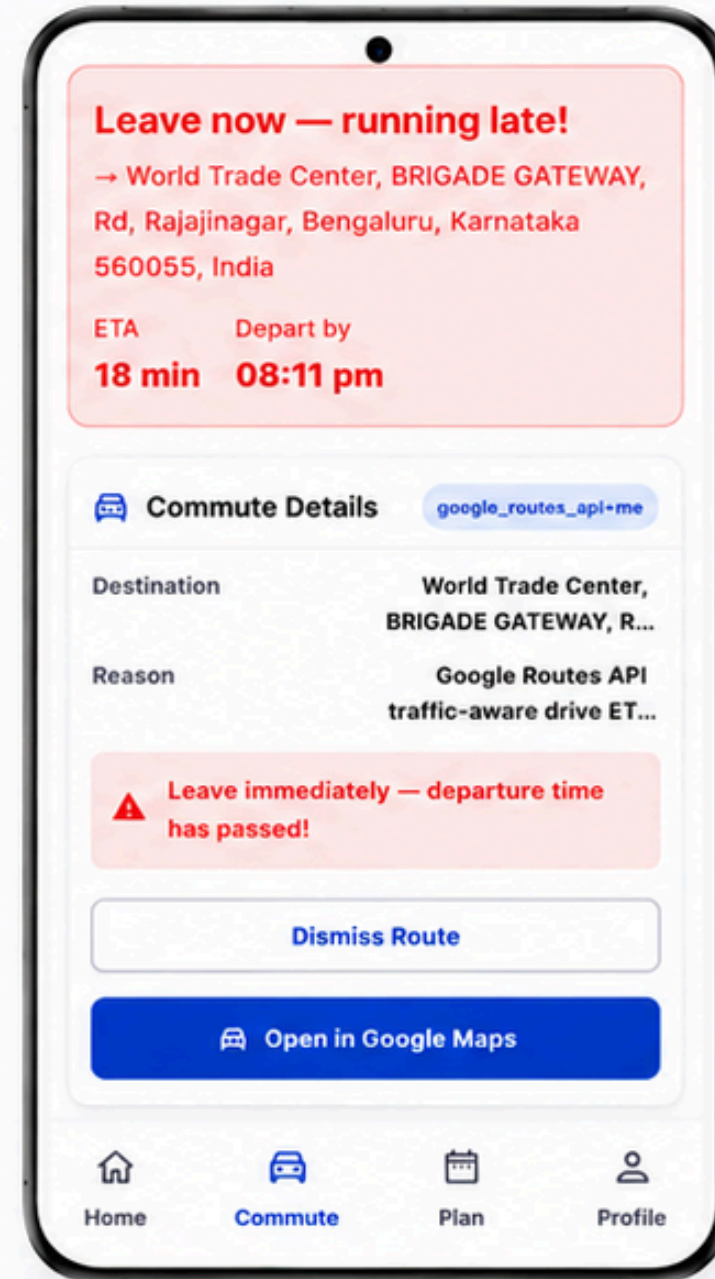
1 Home Dashboard

Real-time stress sensing and adaptive state monitoring



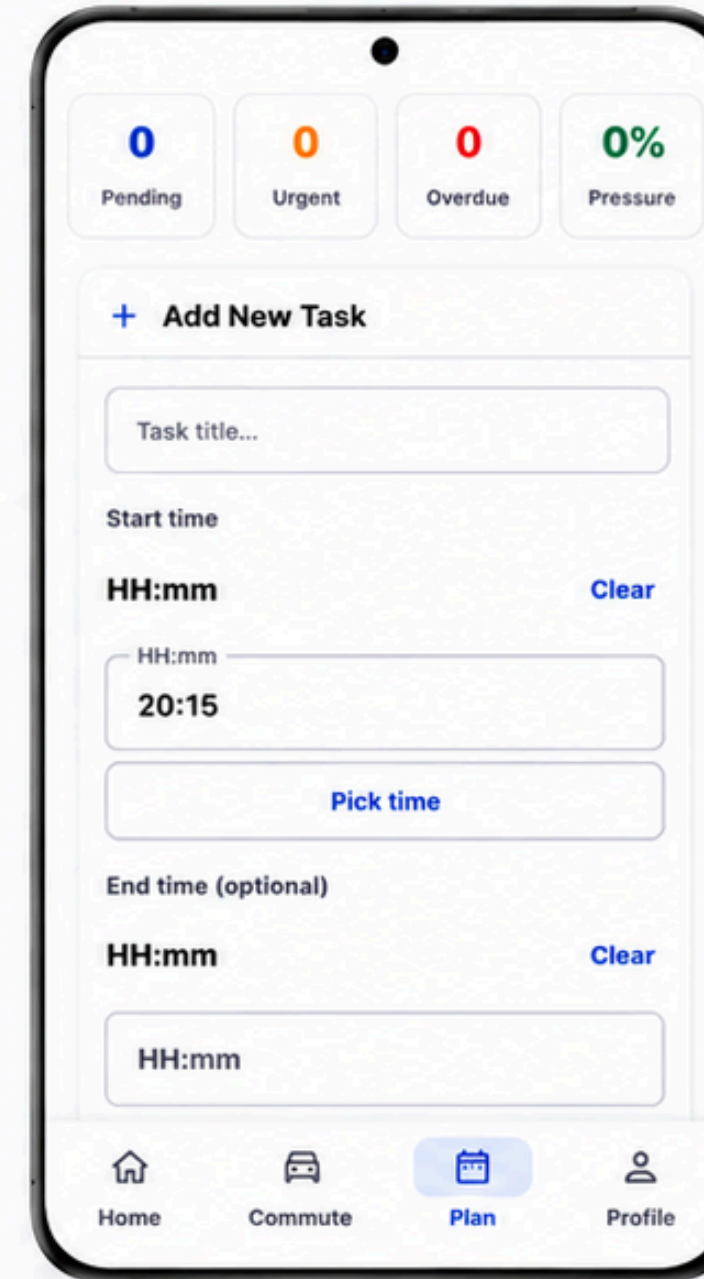
2 Commute Alert

Traffic-aware urgency detection with proactive intervention



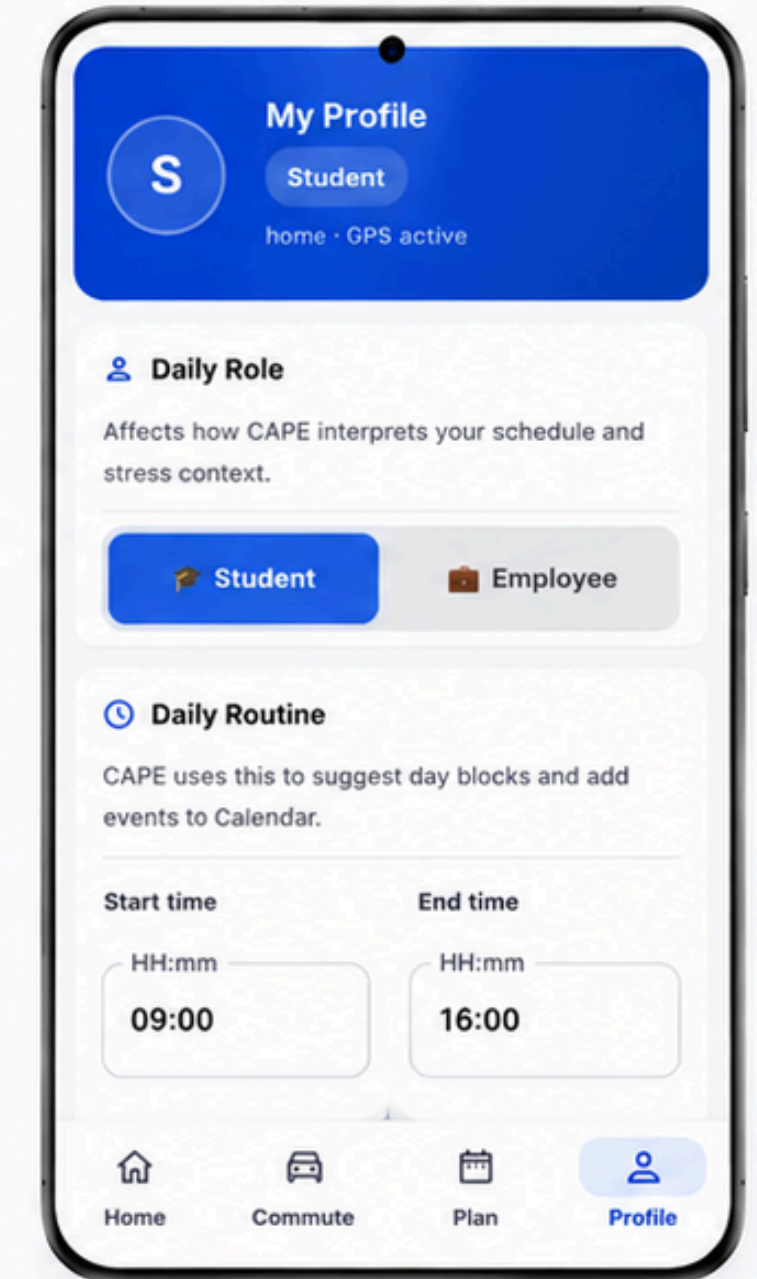
3 Smart Planner

Context-aware task orchestration and pressure balancing



4 Personalization Setup

Saved places, routines and adaptive policy preferences



Stress-aware AI

Continuously monitors and adapts to your mental state.



Commute Orchestration

Real-time traffic intelligence with proactive alerts.



Explainable Automation

Every action is transparent, traceable and under your control.



Adaptive Memory

Learns your patterns to personalize routines and recommendations.

Tech & Architecture

Android Layer

Kotlin + Jetpack Compose

Sensors

calendar, screen state, app behavior, notifications and location signals are captured with permission scope.

UI

Compose dashboard shows stress, active pack, upcoming context and explanation trace.

Actions

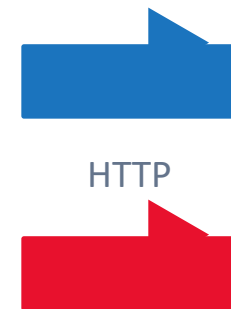
Executes phone-safe controls such as DND, ringer, brightness, reminders and departure alerts.

Fallback

If gateway is unreachable, conservative local rules preserve core safety behavior.

Permission broker

Central Android boundary that prevents agents from acting outside user-granted capabilities.



Gateway Layer

Node.js + OpenClaw orchestration

Ingress

Normalizes Android events into a compact context packet for the agent runtime.

Policy

Selects behavior pack by scoring stress, commute risk, meeting urgency and user preference.

Safety

Confidence gate blocks low-certainty actions and routes them to user confirmation.

Explain

Stores human-readable rationale so automation remains transparent and judge-demo friendly.

Feedback loop

Override → JSONL memory → routine weights → next decision. This is the adaptive core.

OpenClaw Multi-Agent Runtime

Context Intake

merges signals

Routine Memory

retrieves patterns

Stress Score

weighted signals

Commute

ETA + buffer

Orchestrator

ranks actions

Safety

permission gate

Pack Exec

phone actions

Learning

override loop

Google Maps API

YAML / JSONL memory

Impact & Use Cases

Student exam mode

Problem: study sessions are broken by casual app alerts. CAPE response: detect study block + switching, activate focus pack with urgent exceptions. Outcome: fewer breaks and calmer revision.

Office commute stress

Problem: meeting pressure rises when ETA changes late. CAPE response: compute leave-by time, route risk and buffer. Outcome: earlier action and lower last-minute stress.

Deep work session

Problem: users manually defend focus against notification bursts. CAPE response: batch reminders, apply DND and explain automation. Outcome: longer uninterrupted focus window.

Evening recovery

Problem: high screen intensity continues after work. CAPE response: detect fatigue proxies and switch to recovery_mode. Outcome: softer phone state and better personal time.

-30%

manual switching: fewer user-initiated setting changes

+18%

focus retention: longer protected blocks

+22%

on-time starts: better departure timing

-25%

alert fatigue: fewer low-value interruptions

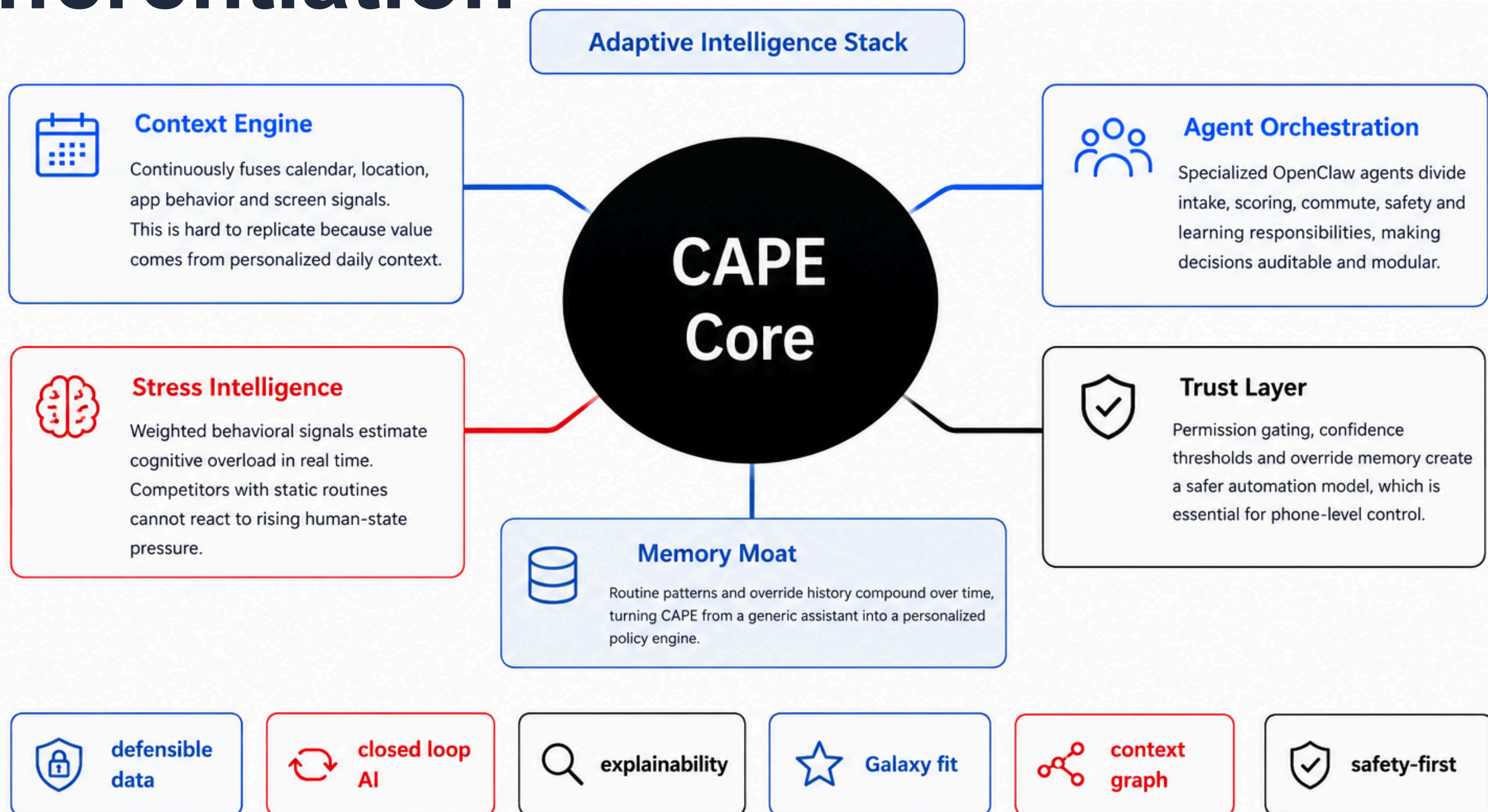
Measurement plan

A/B observe-only baseline
pack acceptance rate
override frequency
commute alert accuracy

Before vs after

Before: reactive phone + manual modes + scattered attention → After: context-aware orchestration + explainable action + adaptive memory

Differentiation



Demo Link



GitHub Repository

Source code, issues,
documentation & more.

[Github Repo Link](#)



Demo Video

Watch the full walkthrough
and key use cases.

[Youtube video link](#)

Brownie Slide

